What Is a Mechanism?

A Five-Axis Ontology for Mechanistic Interpretability

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Abstract

Mechanistic interpretability has produced circuits, features, causal subspaces, and activation-steered algorithms—each resting on an implicit answer to a prior question: what kind of thing is a mechanism? These are not terminological variants. They differ in ontology, and ontological differences propagate: what a mechanism *is* determines when two descriptions pick out the *same* one, what *evidence* can warrant a claim, and what *inferences* that claim licenses. Many apparent disagreements in the literature are ontological disagreements being conducted at the wrong level.

This paper introduces *mechanistic views*: explicit background accounts of what mechanisms are, how they are individuated, and what evidence can support claims about them. A view is a 5-tuple $(O, \sim, E, F, T)$—an ontology, an identity criterion, an evidential standard, a formalism, and an explanatory target. We give coherence conditions on this structure and show that incoherent views produce unresolvable disagreements between evidence and identity: disagreements that no additional experiment can settle, because each new measurement either agrees with the declared identity criterion (adding nothing) or conflicts with it (adding another contradiction).

The five components are not independent. Ontology determines identity, which determines formalism—a chain that explains why most coherence violations are mismatches between the first three axes. We present an atlas of eight views currently operative in the literature, a five-axis classification of 15 common interpretability methods by their implicit view commitments, and four worked examples showing what each view claims and what evidence each leaves open.

The framework is a diagnostic tool. Making the view explicit lets researchers state claims at the level their evidence supports, locate disagreements at their source, and design experiments that discriminate views rather than confirm them.

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1 Introduction

Mechanistic interpretability aims to explain what trained neural networks are computing. The field has made this aim concrete: transformer circuits provide a component-level vocabulary for attention heads and MLPs [Elhage et al., 2021]; induction heads account for a mechanism involved in in-context learning [Olsson et al., 2022]; the indirect object identification (IOI) circuit names the attention heads involved in a specific syntactic task [Wang et al., 2022]; sparse autoencoders recover interpretable directions from activation spaces [Bricken et al., 2023, Templeton et al., 2024]; causal abstraction methods test whether high-level variables are realized in distributed representations [Geiger et al., 2021, 2024]; and weight-space analyses recover inter-layer communication channels and circuit structure from weight matrices alone [Merullo et al., 2024, Elhage et al., 2021].

Yet the word *mechanism* does not mean the same thing in all of these cases. In one paper a mechanism is a set of attention heads; in another it is a functional role such as "name mover"; in another it is a causal subspace recovered by distributed alignment search; in another it is a training-time transition. These are not merely terminological differences. They carry different answers to five basic questions:

(1) **Ontology**: what kind of entity counts as a mechanism?

(2) **Identity**: when do two descriptions refer to the same mechanism?

(3) **Evidence**: what measurements can warrant a claim about a mechanism?

(4) **Formalism**: what mathematical language expresses the claim?

(5) **Target**: what phenomenon is the mechanism supposed to explain?

These questions are not new. The philosophy of science has studied mechanisms for decades: the Machamer-Darden-Craver (MDC) account defines mechanisms as entities and activities organized to produce a phenomenon [Machamer et al., 2000]; Glennan argues that mechanisms ground causal claims [Glennan, 1996]; Craver develops mutual manipulability as the criterion for constitutive relevance [Craver, 2007]; Bechtel and Richardson study decomposition and localization as discovery strategies [Bechtel & Richardson, 1993]; and Illari and Williamson survey the diversity of mechanism concepts across the sciences [Illari & Williamson, 2012]. What interpretability adds is precision about the objects: neural network mechanisms are computationally specified, fully inspectable, and amenable to exact intervention. What interpretability lacks is the conceptual infrastructure that the philosophy of mechanisms provides. This paper bridges the gap.

Most published work answers these questions implicitly. The answers are recoverable from the methods used and the conclusions drawn, but they are not stated—and because they are not stated, disagreements that are fundamentally ontological are argued as disagreements about experimental design.

Three motivating cases. Underdetermination. Activation patching establishes that a head is causally relevant to a behavior. Does this establish a mechanism? Under the object view, perhaps yes. Under the role view, only if the functional role is independently specified and tested. Under the subspace view, not unless the relevant variable is localized to a specific causal subspace. Under the structural view, not unless the result is invariant under the relevant parameter symmetries. A single activation-patching result means different things under different views, and the disagreement cannot be resolved by further patching experiments alone—because all such experiments are equally consistent with multiple views.

Cross-model comparison. The claim that GPT-2 Small and Pythia-160M implement the same induction head mechanism requires a notion of mechanism identity that is cross-model. Component indices are trivially not cross-model. Functional roles may be, if independently specified. Gauge-invariant subspace structure may be, if a canonical transport map is defined. Training-time trajectories may be, if the dynamical basins correspond. Which notion is operative determines whether "the same mechanism" is meaningful across architectures—and therefore whether cross-model generalization claims are well-formed.

Safety-relevant inference. Safety work [Ngo et al., 2022] reasons from "circuit X is present" to "the model has goal Y" or "the model would behave Z under distribution shift." These inferences require different things depending on what a mechanism is. Under an object view, circuit presence is a fact about specific components in a specific model on a specific distribution. Under a structural view, a circuit present as a gauge-invariant structure is robust to reparameterization. Under the instrumental view, a circuit label is warranted by behavioral predictions. The inference from circuit presence to safety-relevant conclusion requires specifying which of these is operative.

Contributions. This paper makes three contributions:

1. A formal definition of a *mechanistic view* as a five-component structure $(O, \sim, E, F, T)$ with typed components and coherence conditions (§2).

2. An atlas of eight views currently operative in the interpretability literature, with full axis assignments, evidence standards, discriminating experiments, and failure modes (§3).

3. The determination chain—ontology implies identity implies formalism—with a five-axis methods mapping showing how common tools carry implicit view commitments (§3.9, §3.11).

These are applied through four worked examples (§4), five recurring decision points (§5), and preliminary cross-view promotion conditions (§6).

Relationship to Marr's levels. The most well-known framework for levels of analysis in computational science is Marr's tri-level account [Marr, 1982]: a computational level (what is computed and why), an algorithmic level (what representation and process), and an implementational level (how the algorithm is physically realized). Marr's levels organize *explanatory targets*—they ask what a system does, how it does it, and what substrate realizes the process. The five axes here organize *ontological commitments*—they ask what kind of object the mechanism is, when two are the same, and what evidence can support the claim. Marr's levels do not distinguish views: two researchers at the same Marr level can hold different mechanistic views (object vs. subspace vs. structural). The eight views are complementary to Marr's hierarchy, not a refinement of it: both are needed, and neither subsumes the other.

2 Mechanistic Views

2.1 Formal Definition

A mechanistic view is a background commitment that a researcher makes, explicitly or implicitly, when they ask what a mechanism is. We give a formal definition, then unpack each component.

Definition 2.1 (Mechanistic view)**.** *A mechanistic view σ is a tuple $(O, \sim, E, F, T)$ where:*

*(i) O is a set, the **ontology**, whose elements are the entities that count as mechanisms under σ.*

*(ii) $\sim \subseteq O \times O$ is an equivalence relation on O, the **identity criterion**: $x \sim y$ means descriptions x and y refer to the same mechanism under σ.*

*(iii) E is a set of **evidential standards**: pairs (m, C) where m is a measurement type and $C \subseteq 2^O/\sim$ is the set of claims about equivalence classes of mechanisms that m can in principle support.*

*(iv) F is a **formalism**: a mathematical or representational language in which elements of O can be expressed and $\sim$ can be computed.*

*(v) T is a **target**: a specification of the phenomena the view is designed to explain.*

Several features of Definition 2.1 deserve comment.

The identity criterion $\sim$ is required to be an equivalence relation (reflexive, symmetric, transitive). This is not automatic: some informal notions of mechanism identity in the literature are not transitive. For example, saying mechanisms A and B are "related" by shared components, and B and C are "related" by shared function, does not imply A and C are the same mechanism. Making $\sim$ an equivalence relation forces precision.

The evidential standard E is explicitly typed: each measurement type m is associated with the claims C it can support. This rules out the common practice of using a measurement to support a claim that is not in the range of that measurement. Activation patching establishes causal relevance of a component; it is not in the range of subspace localization claims unless supplemented.

Definition 2.2 (Coherence)**.** *A view $\sigma = (O, \sim, E, F, T)$ is* coherent *if:*

*(i) **Identity is grounded**: $\sim$ is definable using only the objects in O and the language of F. (A view whose identity criterion involves objects not in its own ontology is incoherent.)*

*(ii) **Evidence tracks identity**: for every measurement type $m \in E$, the claims $C(m)$ are expressible in terms of $\sim$-classes, not finer-grained distinctions. (Evidence that distinguishes objects within a $\sim$-class is picking up on a different ontology than the one declared.)*

*(iii) **Formalism is expressive**: every element of O has a representation in F, and $\sim$ is computable in F.*

Observation 2.1 (Incoherent views produce contradictory demands). Let $\sigma = (O, \sim, E, F, T)$ be a view that violates coherence condition (ii): there exists a measurement type $m \in E$ and objects $x, y \in O$ with $x \sim y$ but $m(x) \neq m(y)$. Then there exist mechanistic claims that are simultaneously required by $\sim$ to be true (since x and y are the same mechanism) and required by E to be false (since m distinguishes them). No experiment can satisfy both demands. This is immediate from the definitions: the claim $\phi =$ "x and y are the same mechanism" is asserted by $\sim$ and refuted by m, so the view entails $\phi \wedge \neg\phi$.

A stronger consequence follows: incoherent views are not merely contradictory in principle but produce systematic evidential underdetermination in practice.

Observation 2.2 (Incoherent views produce systematic underdetermination). Let $\sigma = (O, \sim, E, F, T)$ violate coherence condition (ii), so there exist $x \sim y$ in O with measurement $m \in E$ such that $m(x) \neq m(y)$. Then for every finite evidence set $\{m_1, \ldots, m_k\} \subseteq E$ that includes m, there exist mechanism hypotheses H_1 and H_2 such that: (a) H_1 and H_2 are identical under $\sim$ (same mechanism), (b) H_1 and H_2 are distinguishable by the evidence set, and (c) no additional evidence from E can resolve whether the difference is real or an artifact of the mismatch. This follows by taking $H_1 = H_2 = [x]_\sim = [y]_\sim$ and observing that any further measurement $m' \in E$ either agrees with $\sim$ (offering no information about the m-discrepancy) or disagrees (introducing its own underdetermination).

Proposition 2.2 is the practical payoff: incoherent views do not merely produce logical contradictions—they produce *unresolvable* disagreements between evidence and identity, where each new measurement either agrees with the identity criterion (offering no information about the discrepancy) or disagrees (adding another unresolved conflict). This is the formal version of the empirical observation that debates between, say, activation-patching advocates and DAS advocates cannot be settled by more of the same evidence when the participants hold different implicit views.

Propositions 2.1 and 2.2 are useful diagnostically. A common pattern in interpretability is to declare that mechanisms are gauge-invariant structures (structural view ontology) but then to use activation-patching results as evidence of mechanism presence without checking gauge invariance. If the patching result changes under a reparameterization that preserves computation, it is distinguishing objects that the structural view says are identical—and by Proposition 2.2, no amount of additional activation-patching evidence can resolve this mismatch.

2.2 The Five Axes

We organize the components of Definition 2.1 into five named axes. The axes are not independent: as coherence requires, choices on one axis constrain the others.

Ontology (O). What kind of entity counts as a mechanism [Quine, 1948]? Options include: concrete model components (heads, neurons, MLP sublayers, SAE dictionary elements), functional roles (descriptions of what a component does independently of which component does it), causal subspaces (linear or nonlinear submanifolds of the residual stream that mediate a causal variable), gauge-invariant structures (equivalence classes of weight configurations under symmetries that preserve the computation), temporally extended processes (formation pathways or execution trajectories), points in a stratified mechanism space, measurement perspectives, and predictive models.

Identity ($\sim$). When do two mechanism descriptions denote the same mechanism? Options include: component overlap ($x \sim y$ iff they name the same set of heads or neurons), role equivalence ($x \sim y$ iff they specify the same functional input–output mapping), subspace proximity ($x \sim y$ iff the principal angles between the causal subspaces are all below a threshold θ, i.e. geodesic distance on $\mathrm{Gr}(k, d)$ is small), gauge-orbit membership ($x \sim y$ iff there exists a computation-preserving symmetry g with $g \cdot x = y$), and dynamical-basin membership ($x \sim y$ iff the formation trajectories converge to the same attractor).

Evidence (E). What measurements can support what claims? Key distinctions: activation patching establishes causal relevance of components under a fixed distribution; DAS/IIA with linear alignment constraints measures causal subspace identity; weight composition scores measure structural information flow independent of any input distribution; checkpoint analysis establishes formation dynamics; multi-method robustness tests perspectival convergence. The formal epistemology of evidence accumulation [Earman, 1992, Howson & Urbach, 1989] suggests that independent evidence converges faster than correlated evidence—a principle that grounds the triangulation requirement in the perspectival view.

Formalism (F). What representational language is appropriate? Directed graphs (circuits); functional decompositions (roles); Grassmannian geometry $\mathrm{Gr}(k, d)$ (subspaces); fiber bundles and gauge groups (structural view); dynamical systems and phase portraits (process view); Whitney stratifications (stratified view); measurement algebras (perspectival view).

Target (T). What is the mechanism supposed to explain? A specific behavioral output on a specific distribution; a functional class across models; a representational variable across inputs; a computation class across parameterizations; a formation event; a resolution-relative phenomenon; or a behavioral prediction.

3 An Atlas of Views

We identify eight views currently operative in the literature. Table 1 gives the full axis assignments; Figure 1 shows how the five axes organize the eight views. The views are not mutually exclusive: a single paper may use several, and convergence across views constitutes stronger evidence than any single view. The point is to distinguish them so that each view's commitments and limitations can be assessed independently.

3.1 Object View

The object view treats mechanisms as concrete model components: specific attention heads, neurons, MLP sublayers, or sparse autoencoder dictionary elements. It is the implicit view in circuit diagrams. The IOI circuit as reported by Wang et al. [2022]—26 attention heads organized into functional groups—is an object-level description: the claim is about those specific computational units.

Identity. Two circuit descriptions refer to the same mechanism iff they name the same components. This is precise within a model but trivially fails cross-model: a head in GPT-2 Small is not an object in Pythia-160M.

View	**Ontology (O)**	**Identity ($\sim$)**	**Evidence (E)**	**Formalism (F)**	**Target (T)**
Object	Concrete part: head, neuron, feature	Component overlap	Ablation, activation patching	Directed graph	Specific behavior
Role	Functional role in computation	Role equivalence (same I/O mapping)	Role-specific causal tests, cross-model	Functional decomposition	Functional class
Subspace	Causal subspace of residual stream	Same projector; $d_{\mathrm{Gr}} < \theta$	DAS/IIA (linear), subspace stability	$\mathrm{Gr}(k, d)$	Representational variable
Structural	Gauge-invariant relational structure	Gauge-orbit membership	Holonomy, comp. scores, reparameterization	Fiber bundle quotient	Computation class
Process	Formation trajectory or execution regime	Same trajectory type or dynamical basin	Training checkpoints, formation knockouts	Dynamical system	Mechanism origin
Stratified	Point in stratified mechanism space	Same stratum + local equivalence	Dimensionality tests, localization diagnostics	Whitney stratification	Resolution-relative phenomenon
Perspectival	Projection of mechanism onto context	Cross-method coherence	Multi-method robustness	Measurement algebra	Method-relative observation
Instrumental	Useful predictive or interventional model	Predictive equivalence	Forecast accuracy, intervention utility	Model theory	Behavioral prediction

Table 1: The eight mechanistic views with full axis assignments.

Evidence. Ablation and activation patching are the natural evidence types. Removing a component's activation and observing behavioral degradation establishes causal necessity; replacing it with a counterfactual activation establishes causal sufficiency relative to that counterfactual. The method is strongest when the component set is stable across prompt distributions and the ablation effect is large and specific.

Failure modes. *Role inflation*: a component is identified by patching, then labeled with a role ("name mover," "induction head") that implies richer mechanistic content than the object-level evidence supports. The component identity is established; the role is a hypothesis. *Distribution sensitivity*: the circuit identified by patching on one distribution may differ substantially on another, meaning the claimed object-level mechanism is distribution-relative.

Discriminating experiment. To test whether the object view is appropriate: ablate the identified component set and test whether behavioral degradation is (a) specific to the behavior the mechanism is supposed to implement and (b) stable across prompt distributions. If (b) fails, the object-level description is distribution-relative and a role or subspace description may be more appropriate.

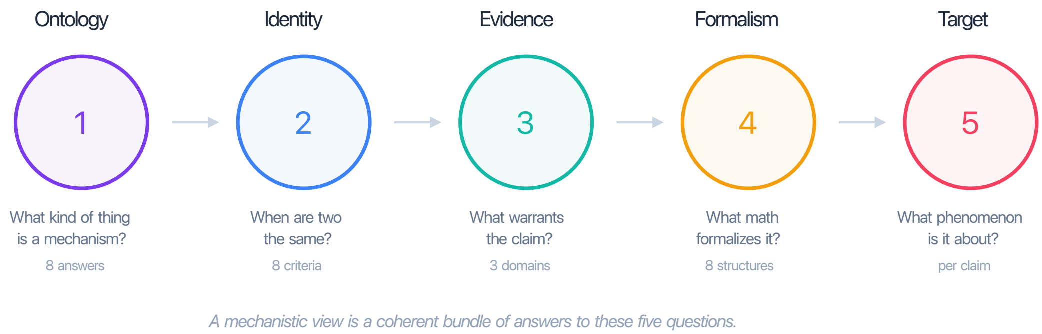

Figure 1: The five axes of a mechanistic view correspond to the five components of Definition 2.1: $(O, \sim, E, F, T)$. A mechanistic view is a coherent bundle of answers to these five questions.

3.2 Role View

The role view treats mechanisms as functional roles that can be realized by different components—the role is a universal that can be multiply instantiated [Lewis, 1986]. “Name mover” is a role: it describes what a component does (copy a name token to the output position), not which head does it. The role view explains why functionally equivalent components across different models can be meaningfully compared.

Identity. Same functional input–output role, possibly in different components or models. Role identity requires a specification of the role that is independent of the discovery procedure (i.e., not merely “does what this head does”) and tests that go beyond the ablation that revealed the component.

Evidence. Role-specific predictions that are not entailed by mere causal relevance. For a name-mover role claim: the weight-space signature of W_{OV} should encode a copying operation; the head should perform this function on novel prompts; a head in a different model with the same W_{OV} signature should earn the same label. The object-to-role slide is a common source of overclaiming: establishing that head 9.9 is causally necessary for IOI does not establish that it is a name mover—the ablation evidence supports the object claim (this component matters) but not the role claim (this component performs a specific function) until the role is independently operationalized and tested.

Failure modes. *Role proliferation*: a new role label is coined for each component discovered, re-describing the data without adding explanatory content. *Underdefined roles*: role labels are informal enough to be true of any component, making them unfalsifiable.

Discriminating experiment. Take the role specification and apply it to a model in which the original component has been ablated. If the role is real, either another component should take over (backup mechanism) or the behavior should degrade in a specific role-consistent way. If no such prediction is confirmed, the role label has not been independently validated.

3.3 Subspace View

The subspace view treats mechanisms as low-dimensional causal subspaces of the residual stream. The key observation is that distributed alignment search [Geiger et al., 2021, 2024] identifies not a component but a rotation of the activation space: the causal variable is encoded in a subspace that may span many components. If alignment matrix Q and its rotation QR both achieve the same interchange intervention accuracy (IIA), the relevant object is the subspace $\mathrm{span}(Q)$, not the matrix Q.

Identity. Two subspace mechanisms are the same iff they project onto the same subspace of $\mathbb{R}^d$: same projector $QQ^\top$, or equivalently small geodesic distance on $\mathrm{Gr}(k, d)$. This is basis-invariant and can be meaningful cross-model when corresponding residual streams can be aligned.

Evidence. The subspace view requires evidence that a specific subspace mediates a causal variable, not merely that a search over subspaces finds one that works. Native subspace evidence includes: SVD or eigendecomposition of weight matrices recovering the subspace without activation data; Grassmannian distance between subspaces recovered by independent methods (weight vs. activation); and subspace stability across prompt distributions. DAS and IIA are often cited as subspace evidence, but as Table 2 shows, DAS is more precisely a Role-view method with a borrowed subspace parameterization: it validates by intervention success (role equivalence), not by Grassmannian distance. DAS can *discover* a candidate subspace, but confirming that it is the causally relevant subspace—rather than merely a subspace that permits successful interchange—requires convergent evidence from weight-space analysis.

The Sutter constraint. Sutter et al. [2025] showed that unrestricted nonlinear alignment maps achieve 100% IIA even on randomly initialized models, making DAS vacuous without structural constraints on the alignment class. Subspace-view claims therefore require either linear alignment or alignment constrained to respect the transport structure induced by the weight matrices. Subject to this constraint, subspace stability (eigenvalue gap between the causal subspace and its complement) is the natural diagnostic. This constraint recurs throughout the atlas: it limits the evidential range of DAS (Section 3.11), appears in the diagnostic checklist (Appendix A), and is the primary threat to subspace claims in the worked examples.

Failure modes. *SAE confusion*: sparse autoencoder features are dictionary elements that minimize reconstruction loss; causal subspaces are directions that mediate specific causal paths. These are different objectives and can disagree. An SAE feature is a hypothesis about a one-dimensional subspace mechanism, not a confirmed subspace claim. *Sutter vacuousness* (see above): reporting high IIA without specifying the alignment class allows any model to be assigned any causal structure.

Discriminating experiment. Verify that the identified subspace is stable: the subspace recovered by DAS on a held-out distribution should have small geodesic distance from the training-distribution subspace. Also confirm that an aligned basis-change (within the subspace) does not change the IIA, while a rotation that takes vectors out of the subspace does reduce it.

3.4 Structural View

The structural view treats mechanisms as equivalence classes of implementations under transformations that preserve the computation. The relevant object is not a head index, a basis vector, or a specific projection—it is the orbit of implementations related by the symmetry group of the transformer's weight space. Two weight configurations that implement the same computation via different parameterizations are the same mechanism under this view.

The formalism involves gauge orbits, parallel transport maps on the Grassmannian, holonomy groups, and sheaf cohomology—the differential-geometric toolkit of modern gauge theory [Nakahara, 2003].

Identity. Gauge-orbit membership: $x \sim y$ iff there exists a computation-preserving symmetry g (such as a scale transformation, a rotation of a head's query-key subspace paired with an inverse rotation of its output, or a permutation of neurons) such that $g \cdot x = y$. Holonomy fingerprints provide a finer criterion: the holonomy group of the transport map induced by a mechanism's weight matrices is an invariant that does not change under gauge transformations.

Evidence. Gauge-invariant measurements: weight-space composition scores $\|W_{O,u} W_{K,v}\|_F$ measure information flow capacity independently of basis choice [Elhage et al., 2021, Merullo et al., 2024]; holonomy measurements; and reparameterization-stability tests (check that a result survives computation-preserving rescalings of the weights).

Failure modes. *False gauge invariance*: claiming structural equivalence without verifying that the symmetry actually preserves the computation. *Architectural prior confound*: structural features that discriminate trained from randomly initialized models may be detecting architecturally privileged positions rather than learned structure.

Discriminating experiment. Apply a known computation-preserving symmetry (e.g., scale by λ, counter-scale by λ^{-1}) to the weights of a component and retest the activation-space evidence. If the activation-space result changes, the original result was not gauge-invariant and is not a structural-view claim.

3.5 Process View

The process view treats mechanisms as temporally extended. A mechanism is not merely a final-checkpoint artifact but a formation process, an execution trajectory, or a dynamical regime—what Darden calls "reasoning about mechanisms in the making" [Darden, 2006]. This view is natural for phenomena where the how of learning is itself mechanistically relevant: the induction head mechanism, for instance, forms via a phase transition at a specific training step [Olsson et al., 2022]—this is process-level structure that is not recoverable from the final checkpoint alone.

Identity. Same trajectory type or dynamical basin. Two processes are the same mechanism if they follow the same formation pathway, occupy the same attractor, or exhibit the same phase-transition structure. This notion of identity is less developed than component or subspace identity, and formalizing it is an open problem.

Evidence. Checkpoints, gradient flow, loss-curve structure, and formation knockouts. A mechanism claim under the process view cannot be supported by final-checkpoint analysis alone. The induction head formation result provides a clear example: the phase change at a specific training step is visible in the composition score evolution, and this is process-level evidence distinct from any claim about the final circuit [Olsson et al., 2022].

Failure modes. *Process–circuit conflation*: the final-checkpoint circuit and the formation process are distinct objects; the circuit does not determine the process. Two models may arrive at identical final circuits via very different training trajectories. *Training-detail overfitting*: a process described precisely enough to be informative may be specific to a particular optimizer, learning rate schedule, or dataset.

Discriminating experiment. Run the same architecture with different training hyperparameters (learning rate, batch size, optimizer) and compare the formation dynamics. If the final circuit is similar but the formation trajectory differs, the process-level claim is hyperparameter-sensitive and should be stated at the appropriate level of generality.

3.6 Stratified View

The stratified view treats mechanism space as organized into strata: point-localized objects, one-dimensional features, higher-dimensional subspaces, nonlinear manifolds, and fully distributed structures. Rather than forcing a binary choice between localized and distributed, the stratified view asks which stratum the evidence supports and whether the appropriate description changes with measurement resolution.

Figure 2 illustrates the key distinctions schematically. The natural formalism is Whitney stratification [Whitney, 1965, Golubitsky & Guillemin, 1973]: a decomposition of mechanism space into smooth manifold strata that are partially ordered by containment of their closures. The stratified view has a direct analogy in physics: the renormalization group (RG) describes how the effective description of a system changes with measurement resolution [Wilson, 1971, Goldenfeld, 1992]. RG fixed points correspond to stable strata; relevant operators correspond to structure that survives coarse-graining; universality classes correspond to mechanisms that appear identical at a given resolution despite microscopic differences. The stratified view imports this resolution-dependence into interpretability.

A mechanism's stratum is determined by its dimensionality, its localizability in specific components, and the topology of its support in the residual stream.

Identity. Same stratum and same local equivalence class within the stratum. Two descriptions are the same mechanism if they name the same stratum and agree on local structure within that stratum.

Evidence. Dimensionality tests (effective rank of the causal subspace), localization diagnostics (is the mechanism's effect concentrated in a small component set or distributed?), and stratum-transition tests (does the apparent mechanism change stratum when the measurement resolution changes?).

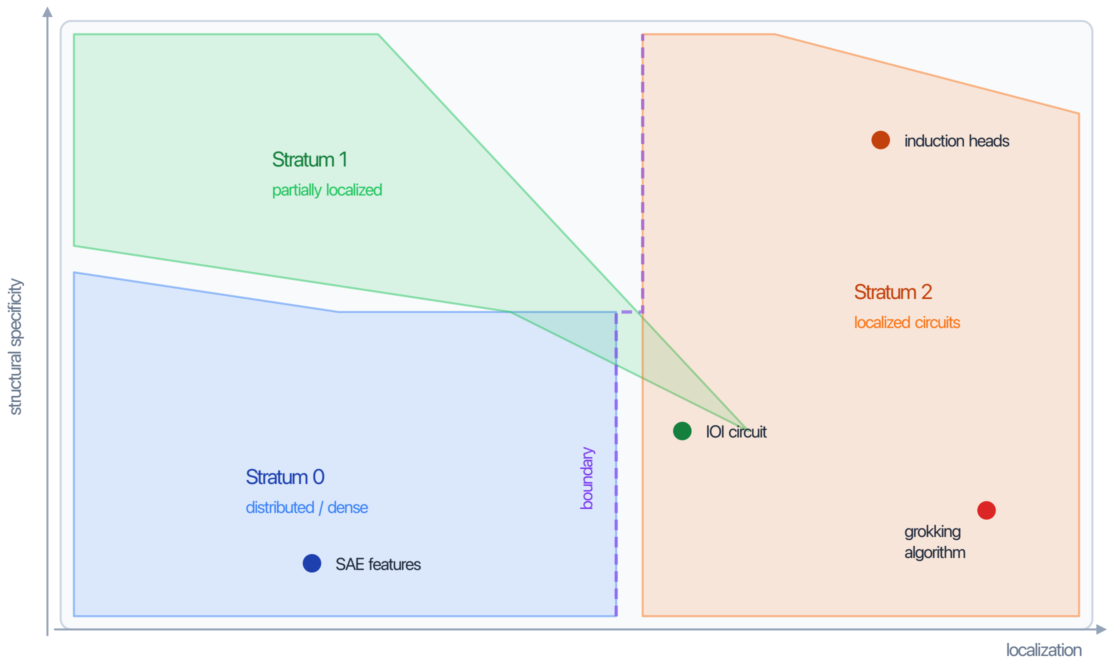

Figure 2: Schematic of mechanism space as a stratified manifold. Each stratum is a smoothly varying subset distinguished by the localization and structural specificity of the mechanism. Boundaries between strata are lower-dimensional singular sets. Empirical examples are placed roughly by the evidence reviewed in Section 4; exact positions are schematic, not quantitative.

Failure modes. *Stratum misidentification from method artifacts.* A genuinely distributed mechanism may appear localized to a single component if the ablation method captures only the dominant direction. Conversely, a localized mechanism may appear distributed if the measurement method has high background noise. The stratified view does not resolve this automatically—it provides the language to state it precisely.

Discriminating experiment. The stratified view predicts that measurement resolution affects the apparent stratum. Test this by comparing the effective dimensionality of the causal subspace recovered by DAS at different hook points and with different alignment constraints. Consistent stratum across resolutions supports a stratum assignment; stratum-change suggests either a higher-stratum mechanism being partially recovered or a measurement artifact.

3.7 Perspectival View

The perspectival view treats methods as revealing partial projections of mechanisms rather than the mechanism in itself. Disagreement between methods is not automatically a failure; it may reveal the method-relative structure of what is being measured. This view is philosophically related to Massimi's perspectival realism [Massimi, 2022]: the claim is not that mechanisms are relative to methods (that would be instrumentalism), but that scientific methods are partial perspectives on a method-independent mechanistic structure. The key distinction—between perspectival *evidence* (measurements are always from a vantage point) and perspectival *truth* (what's true depends on who measures it)—maps directly onto the primary failure mode: conflating the method-relativity of evidence with the claim that mechanisms are themselves method-relative objects.

Identity is coherence across perspectives: two descriptions refer to the same mechanism if they are mutually consistent and jointly interpretable as projections of the same underlying structure. Evidence is multi-method robustness [Tower, 2026], formalized as convergence across structural (weight), causal (activation/intervention), and dynamic (training-time) evidence families. The discriminating experiment is cross-method consistency: test whether a patching-supported claim is also supported by weight-space composition scores. If not, a third method adjudicates.

3.8 Instrumental View

The instrumental view treats mechanisms as useful predictive or interventional models, warranted by forecast accuracy and intervention utility regardless of whether they identify a real internal object. Identity is predictive equivalence: two descriptions are the same mechanism if they make the same predictions about all interventions and behaviors. This has the weakest ontological commitments and is therefore the easiest to satisfy—but also the most limited. The primary failure mode is *safety-relevant overreach*: claims that require strong ontological conclusions ("the model is pursuing a goal") cannot be supported by predictive utility on the training distribution alone. The discriminating experiment: test whether the mechanism label generates novel predictions beyond the behavior used to identify it.

3.9 The Determination Chain: Ontology Implies Identity Implies Formalism

The five axes are not independent. In practice, the relationship is mostly a chain of determination:

$$\text{Ontology} \longrightarrow \text{Identity} \longrightarrow \text{Formalism}$$

Choosing what kind of thing a mechanism *is* determines when two mechanisms are the *same*, which in turn determines what *mathematical structure* can express the claim. If mechanisms are causal subspaces, identity is geodesic proximity on $\mathrm{Gr}(k, d)$, and the formalism must include Grassmannian geometry. If mechanisms are gauge-invariant structures, identity is gauge-orbit membership, and the formalism must include fiber bundles and holonomy groups. The chain is mostly one-to-one: each of the eight ontological commitments in the atlas implies a specific identity criterion, which implies a specific ambient formalism (Table 1).

Evidence and target are constrained by the ontology–identity pair but not fully determined by it: a subspace claim can be supported by activation-space evidence (DAS), weight-space evidence (SVD), or both, and the target is chosen by the researcher. The determination chain therefore has two free parameters once the ontology is fixed.

This structure explains why the most common coherence violations involve mismatches between the first three axes: using a subspace formalism (Grassmannian distance) while holding an object ontology (component overlap identity), or using an object evidence type (activation patching) to support a structural claim (gauge-orbit identity). The chain makes these mismatches visible.

3.10 View Families and Ontological Commitment

The eight views group into three families and two singletons based on their shared mathematical apparatus (Figure 3):

- **Identity family** (Object, Role): mechanisms are identified by what they are (components) or what they do (functions). Lowest mathematical overhead; most of the current literature operates here.
- **Mathematical family** (Subspace, Structural): mechanisms live in geometric spaces and identity is defined by distances or orbits in those spaces. Requires differential geometry.
- **Process family** (Process, Stratified): mechanisms are temporally extended or resolution-dependent. Requires dynamical systems theory or stratification theory.
- **Methodological singleton** (Perspectival): no single method reveals the mechanism; identity requires cross-method coherence.
- **Pragmatic singleton** (Instrumental): mechanisms are useful models; identity is predictive equivalence.

The views can be ordered by increasing ontological commitment—how much they claim about the existence and nature of the mechanism independent of any measurement procedure:

$$\text{Instrumental} < \text{Perspectival} < \text{Object} < \text{Role} < \text{Subspace} < \text{Structural} < \text{Process} < \text{Stratified}$$

Higher commitment views make stronger claims but require more evidence. The instrumental view requires only predictive utility; the stratified view requires evidence across multiple strata and measurement resolutions. Most published interpretability work operates at the Object or Role level. Moving to higher-commitment views—as this paper argues researchers should consider—requires explicitly stating which view is operative and supplying the corresponding evidence.

3.11 Methods Mapping

Each commonly used interpretability method carries implicit commitments across all five axes. Table 2 makes these explicit.

Several patterns are visible. First, the field's default operating point is Object ontology with activation evidence: activation patching, path patching, ACDC, ablation, SAE features, and logit lens all sit here. Second, DAS occupies an interesting intermediate position: it uses a subspace parameterization (the alignment matrix Q searches over subspaces) but validates by interchange intervention accuracy, which is a *role*-level criterion—whether two representations can be swapped without loss of function. The subspace is the search space, not the ontology. Third, Factor EAP is the only method that combines Subspace ontology with both evidence types: the factors define the subspace decomposition (weight evidence), and the attribution scores define which factors are active for a given task (activation evidence). Fourth, no widely-used method operates in the Stratified or Perspectival views—these views generate research programs rather than describing current practice.

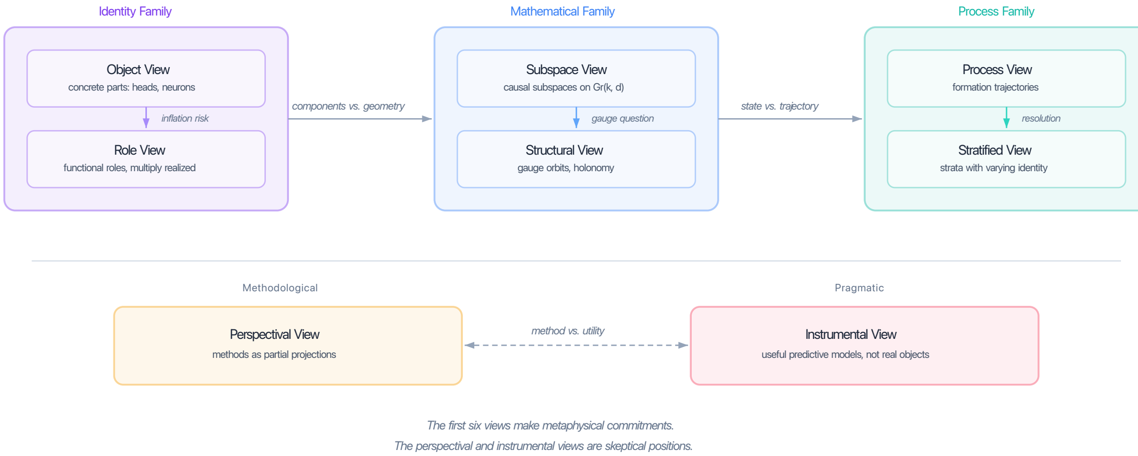

Figure 3: The eight views grouped into three families (identity, mathematical, process) plus two singletons (methodological, pragmatic), ordered by increasing ontological commitment. Within each family, the internal arrow marks the principal decision point between the two views.

4 Worked Examples

Four extended examples show what the atlas does in practice: not just describing views in the abstract, but showing how the same body of evidence is read differently under different views and what each reading implies for follow-up work. The first three (IOI, induction heads, SAE features) illustrate multi-view analysis of final-checkpoint phenomena. The fourth (grokking) illustrates the process view, which cannot be reduced to any final-checkpoint analysis.

4.1 Indirect Object Identification

Wang et al. [2022] identify a circuit for indirect object identification (IOI) in GPT-2 Small: 26 attention heads organized into functional classes including S-inhibition heads, name movers, and duplicate-token detection heads.

Object view. The mechanism is those 26 heads. The evidence (activation and path patching, mean ablation) establishes causal necessity and sufficiency relative to the test distribution. Object-level identity is precise within GPT-2 Small and fails across architectures.

The object-level result is solid. What it does not establish: (a) that these heads perform a functional role that would be recognizable in another model, (b) that the relevant causal variables are localized in specific subspaces, (c) that the result is invariant under parameter reparameterizations, or (d) how the circuit formed. Claims (a)–(d) require distinct evidence.

Role view. The four role classes (S-inhibition, name mover, duplicate-token detection, and backup name mover) constitute a role-level description. This is a hypothesis about what the heads do, not a confirmed role claim. To confirm it, each role should be operationalized and tested independently: name movers should copy tokens in novel syntactic constructions; S-inhibition heads should suppress the subject in novel contexts where the subject–object

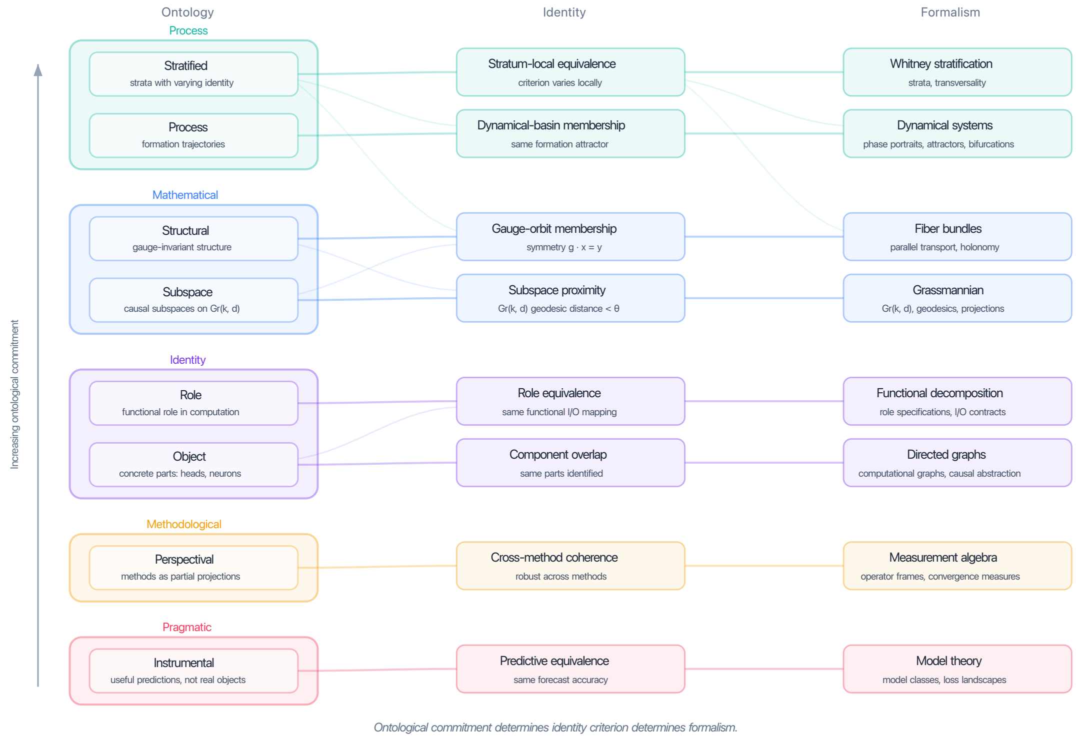

Figure 4: The determination chain: ontology determines identity determines formalism. Each view's ontological commitment (left) implies a specific identity criterion (center), which implies a natural mathematical formalism (right). Primary connections are shown in bold; secondary cross-connections are shown faintly. Increasing ontological commitment runs from bottom to top.

relationship is structurally varied. Until these tests are run, the role labels describe the experimental template more than the mechanism.

Subspace view. The subspace question is where the IO and S tokens, and the inhibitory relationship between them, are represented in the residual stream. DAS with linear alignment constraints would identify the causal subspaces for each variable and test whether they can be independently intervened on. The 26-head object-level description does not resolve this: two different subspace configurations could produce the same patching results.

Structural view. The structural question is which aspects of the circuit are gauge-invariant. Path-patching results that survive computation-preserving reparameterizations of the relevant head weights establish structural facts; those that do not are basis-dependent observations. The holonomy of the circuit's transport map would provide a structural fingerprint that is invariant across model instances.

What each view implies for next steps. Under the object view, the main open question is distribution robustness: does the 26-head circuit generalize to other indirect object constructions beyond the original template? Under the role view, it is role operationalization: what behavioral predictions do the role labels generate beyond the original task? Under the subspace view, it is subspace identification and stability. Under the structural view, it is gauge-invariant characterization via the companion geometric formalism.

4.2 Induction Heads

The induction head result [Olsson et al., 2022] is the strongest mechanistic result in the current literature, precisely because multiple views converge on the same object.

Object view. Induction heads are identified by activation patching. Their distinctive attention pattern—attending to the token immediately after the previous occurrence of the current token—is established across a large family of transformers. Object-level identity is extended cross-model by demonstrating the pattern holds across model sizes and architectures.

Role view. The induction pattern defines a functional role that is operationalized (it can be tested on any sequence with a repeated token) and confirmed across model families. This is one of the few cases in the literature where role identity is genuinely established rather than hypothesized. The role specification is independent of the component: any head that attends to t_j from position i when token $t_i = t_{j-1}$ is an induction head, regardless of layer or position.

Process view. The formation dynamics are studied across training checkpoints. A phase transition is identified at a specific step in the loss curve, and the composition score between the head and its prefix-matching partner increases sharply at that step [Olsson et al., 2022]. This is process-level evidence—it concerns how the mechanism formed, not what it is at the final checkpoint. The formation process is argued to be causally responsible for the in-context learning capacity of the model, a claim that requires process evidence precisely because it is about the training-time origin of a behavior.

What remains open. Under the stratified view: induction heads appear to be point-localized objects, but whether the relevant causal variable is localized to a one-dimensional subspace (object stratum) or spans a small-dimensional subspace is not established. Under the structural view: the holonomy of the induction head transport map has not been characterized. Under the subspace view: the exact subspace of the residual stream encoding the "previous occurrence" variable and the "current token" variable has not been jointly identified via DAS.

The convergence across object, role, and process views is why the induction head result can support relatively strong claims. The gaps in subspace and structural evidence are where further work would strengthen or qualify those claims.

4.3 Sparse Autoencoder Features

Sparse autoencoder (SAE) features recovered from Claude-3 Sonnet [Templeton et al., 2024] and smaller models [Bricken et al., 2023] are semantically interpretable and partially causally validated via steering experiments. The status of SAE features as mechanisms is genuinely contested, and the atlas provides precise vocabulary for the dispute.

Object view. Under the object view, each dictionary element is a potential mechanism: a one-dimensional object in the activation space. The evidence (semantic interpretability, steering effects) supports a weak form of causal relevance. The object-level description is precise: the feature is the specific dictionary vector learned by the SAE.

Subspace view. Each feature is a hypothesis about a one-dimensional causal subspace mechanism. The hypothesis is that the direction of the feature vector corresponds to a direction in the residual stream that mediates a specific causal variable. The SAE training objective (sparse reconstruction) does not guarantee this: the feature might be the dominant activation direction without being the causally relevant one. Confirmation requires linear DAS showing that the feature direction specifically mediates the causal variable in question.

Perspectival view. SAE features are a measurement perspective on the activation space: the perspective defined by the sparse coding objective. This perspective has specific invariances (insensitive to causal structure; sensitive to geometric structure of activations) and specific blind spots (may miss subspaces that are causally relevant but not reconstructively prominent). Under the perspectival view, the right question is not "are SAE features mechanisms?" but "what does the SAE perspective reveal that the weight-space perspective and the causal intervention perspective would independently confirm?" Convergence of all three on the same direction would substantially strengthen the claim.

What each view implies for next steps. Under the object view: test causal necessity via ablation and compare with the steering evidence. Under the subspace view: run linear DAS for the most interpretable features to test whether the feature direction is the causal subspace direction for the relevant variable. Under the perspectival view: compute weight-space composition scores for the same features to identify which ones are also structurally prominent, and compare with the causal evidence.

4.4 Grokking: A Process-View Example

Grokking—the phenomenon where a model suddenly generalizes to the test set long after memorizing the training set—is the clearest case where the process view is not optional. A checkpoint analysis at any single training step shows a model that has either memorized or generalized; the mechanistically interesting phenomenon is the transition between these regimes.

Process view. The mechanism is the formation trajectory itself: the transition from memorization to generalization, the training step at which it occurs, and the circuit-level changes that accompany it. Identity under this view is dynamical-basin membership: two models exhibit the same grokking mechanism if they undergo structurally similar transitions (same sequence of circuit formation events, same loss-curve topology) even if the final circuits differ in detail.

Object view. An object-level analysis of the post-grokking checkpoint identifies the circuit that implements the generalized computation. This is a valid object claim, but it misses the process-level structure entirely: it cannot explain *why* generalization happened, *when* it happened, or whether it would happen under different training conditions.

What each view implies. Under the process view: the key experiments are checkpoint sweeps, formation knockouts (does ablating a component mid-training prevent the transition?), and hyperparameter sensitivity (does the same transition occur with different learning rates?). Under the object view: the key experiment is circuit identification at convergence. The two views answer different questions. Neither subsumes the other, but claims about the *origin* of generalization require process evidence.

5 Five Decision Points

Five recurring forks in empirical interpretability work change in meaning depending on which view is operative. We examine what each view implies for each fork and, where views make conflicting predictions, what evidence would adjudicate.

5.1 Localized vs. Distributed

The question is whether a mechanism is implemented by a small, specific set of components or spread across many.

Under the *object view*, this is an empirical question: which minimal component set is causally necessary and sufficient? Under the *subspace view*, it is about the dimensionality and localizability of the causal subspace: a low-rank subspace localized in a small number of attention heads is one answer; a high-dimensional subspace spanning many heads is another. Under the *stratified view*, it is a question about which stratum the evidence supports.

The views come apart in practice. Activation patching may identify a small object-level component set as causally necessary, while DAS identifies a high-dimensional subspace spanning many components. These need not be contradictory: the patching identifies the components whose activation values lie in the relevant subspace; DAS characterizes the subspace. The stratified view provides the language to say that the two results are compatible descriptions at different levels of resolution.

5.2 Object vs. Role

The question is whether a mechanism claim is about a specific component or about a functional role that component plays. Under the *object view*, the claim is about the component; under the *role view*, it is about the function.

The distinction matters for generalization. An object claim does not entail that any component in any other model is the same mechanism. A role claim does—if the role is operationalized and tested. The slide from object to role happens in one sentence in most papers (“head 9.9, which we call a name mover, ...”): the object identity is established, the role is named, and subsequent sentences treat the role as established without independent validation.

5.3 Single vs. Triangulated Evidence

The question is whether a mechanistic claim requires evidence from a single method or convergent evidence from multiple methods with independent assumptions.

Under the *object view* with activation patching, a single strong patching result may be sufficient. Under the *subspace view*, weight-space and activation-space convergence strengthens the claim (both should identify the same subspace if the view is correct). Under the *perspectival view*, single-method evidence is always vulnerable to that method's confounds: triangulation is definitionally required.

The case for triangulation is not merely a caution. If a method m can only support claims in its range $C(m)$, and the target claim is not in $C(m)$, then no amount of evidence from m can support the claim—however strong that evidence is.

5.4 Static vs. Process

The question is whether the mechanism is a final-checkpoint artifact or a formation process. This is a clean view-level distinction with a clean evidential consequence: process claims need process evidence; checkpoint claims cannot support process conclusions.

This matters most for: (a) training-dynamics hypotheses ("the model learned to do X because ..."), (b) phase-transition-like generalization patterns, and (c) claims about the origin of in-context learning behaviors. In all three cases, checkpoint evidence establishes what the mechanism is at convergence but not how it came to be.

5.5 Subspace vs. Structural

The question is whether mechanism identity is determined by the subspace a mechanism occupies or by the gauge-invariant structure of the computation it performs. A subspace claim is basis-invariant: the subspace $\mathrm{span}(Q)$ is the same regardless of which orthonormal basis of Q is used. But it is not symmetry-invariant: two subspace mechanisms in the same gauge orbit (related by a computation-preserving symmetry that rotates the subspace) would be the same mechanism under the structural view but potentially different under the subspace view.

The structural view is more realist and harder to operationalize. The subspace view is more tractable and directly supported by current DAS methodology. For most practical questions in the field, the subspace view is appropriate; the structural view becomes important when comparing across parameterizations of the same architecture or when making claims that should be invariant to training randomness.

5.6 Behavioral vs. Mechanistic

The question is whether a finding is about a model's *behavioral propensity* (how it tends to behave on a distribution) or about a *mechanism* (an internal structure that produces behavior). Steinhardt [2026] argues that the field over-invests in capability evaluation (can the model do X?) and under-invests in propensity evaluation (does the model typically do X, and under what conditions?).

Under the *instrumental view*, a behavioral finding—the model reliably outputs Y given X—is sufficient. A propensity is what you care about, and a mechanism is irrelevant. Under the *object view*, a behavioral finding is a starting point: the question becomes which components produce that behavior, and whether they are necessary. A capability finding (the model *can* do X) is weaker than a propensity finding (the model *typically does* X) because the former requires only that the relevant mechanism exists, while the latter requires that it be routinely recruited.

The distinction matters most for safety. A steering vector that suppresses refusal behavior on a test distribution establishes an instrumental capability—not a mechanistic understanding of refusal [Turner et al., 2024]. Whether that vector *is* the refusal mechanism, or merely interferes with it, requires at least object-view evidence (ablation of specific components) and ideally subspace-view evidence (stability of the intervention direction across distributions). The gap between instrumental evidence and the safety conclusions drawn from it is the central evidence deficit identified by the open-problems analysis in Appendix C.

6 Practical Implications

Stating the view is a publication norm. The primary practical implication of this paper is that mechanistic interpretability claims should state the view under which they are made. This is not bureaucratic overhead—it is a precision requirement of the same kind as reporting which distribution patching was run on. Appendix A provides a diagnostic checklist for identifying coherence violations in published and in-progress work. Object-level claims should be labeled as such, with the relevant distribution specified. Role claims should specify the role operationally and report the independent tests. Subspace claims should report the identity criterion (same projector? geodesic distance below what threshold?) and the subspace stability. Structural claims should report which symmetries were tested. Process claims should report the temporal evidence.

Evidence-claim matching. Each measurement type has a range: the set of claim types it can support. A measurement whose range does not include the target claim type cannot support that claim, regardless of how strong the measurement is. Activation patching is not in the range of structural claims. Single-method IIA without alignment constraint is not in the range of any claim. DAS with linear alignment is in the range of one-dimensional subspace claims. Stating the view makes these range requirements explicit and checkable.

Independent convergence. Williams et al. [2025] argue from philosophy of science that MI needs philosophical foundations—for clarifying concepts, refining methods, and navigating epistemic complexity. Their three central examples (decomposition identity, feature ontology, deception detection) map directly onto view-level analyses in this paper: "no single correct decomposition" corresponds to the object–subspace distinction (Section 2.2); "vehicle vs. content" corresponds to the object–role split applied to features; and "behavioral detection is insufficient for deception" corresponds to the instrumental view's ceiling (Appendix C). The convergence from independent starting points—philosophy of science vs. measurement theory—strengthens both analyses and suggests that the underlying structure is not an artifact of either framing.

Views as research programs. The object and role views primarily organize and classify existing empirical practice. The subspace, structural, and stratified views generate positive research programs: each calls for methods, results, and formalism that do not yet exist. For the subspace view: systematic DAS studies with stability tests across models and distributions. For the structural view: holonomy measurements, gauge-invariance checks, and the development of the sheaf cohomology formalism. For the stratified view: dimensionality diagnostics and stratum-transition tests. The atlas is therefore not merely a taxonomy of current practice; it is a map of what is missing.

7 Cross-View Promotion

Promotion conditions. A common question is: under what conditions can a result at one view be promoted to a higher-commitment view? The following are preliminary promotion conditions:

- **Object → Role.** The component must be tested with an independently operationalized role specification on held-out constructions. If the role predicts behavior beyond the original discovery distribution, the result can be stated as a role claim.
- **Role → Subspace.** The functional role must be shown to be mediated by a specific low-dimensional subspace (e.g., via linear DAS with the Sutter constraint), and this subspace must be stable across prompt distributions with small geodesic distance on $\mathrm{Gr}(k, d)$.
- **Subspace → Structural.** The subspace must survive computation-preserving symmetries of the weight space. If the subspace recovered by DAS changes under a reparameterization that does not change the model's computation, the subspace claim is not structural. Holonomy measurements provide a gauge-invariant fingerprint.
- **Any → Process.** The static claim must be supplemented with training-dynamics evidence: checkpoint analysis showing when and how the mechanism formed, formation knockouts, or loss-curve structure. No amount of final-checkpoint evidence can substitute.

Each promotion requires new evidence of the kind specified by the target view. Evidence from the source view does not accumulate toward the target view's standard—it answers a different question. The chain is not sequential: promotion can skip levels when the evidence supports it. For example, an Object-level result can promote directly to Subspace if linear DAS identifies a causal subspace spanning the relevant components, without requiring an intermediate Role specification. The subspace evidence subsumes the object evidence (the components are where the subspace has support) but does not require role-level mediation.

View selection: a practical heuristic. The framework is pluralist, but a researcher facing a concrete question needs guidance on which view to adopt. The following heuristic matches the scope of the claim to the minimum view that can support it:

- If the claim is about *a specific model on a specific distribution*, the **Object view** is sufficient.
- If the claim should *generalize across models* ("Pythia and GPT-2 both have this mechanism"), at least the **Role view** is required—component indices are trivially not cross-model.
- If the claim involves a *distributed representation* not localized to a single component, the **Subspace view** is needed.
- If the claim should *survive reparameterization* (permutation, rescaling, rotation of weight matrices), the **Structural view** is needed.
- If the claim is about *how the mechanism formed* or why it appeared at a specific training step, the **Process view** is needed—no amount of checkpoint evidence can substitute.
- If the mechanism's *apparent type changes with measurement resolution*, the **Stratified view** is the appropriate language.

The heuristic is conservative: it recommends the lowest-commitment view that matches the claim's scope. Higher-commitment views are always available but require correspondingly stronger evidence.

Connections to neighboring fields. The five-axis framework has natural translations into several disciplines whose concepts it draws on.

Philosophy of science. The ontology axis corresponds to the central question of the new mechanist literature: what counts as a mechanism [Machamer et al., 2000, Craver, 2007, Illari & Williamson, 2012, Glennan, 1996]? The identity axis operationalizes the question that Lewis's possible-worlds framework [Lewis, 1986] raises for functional kinds: under what counterfactual conditions does a mechanism remain "the same"? The evidence axis inherits the eliminativist structure of Earman's Bayesian epistemology [Earman, 1992] and the domain-triangulation principle of Howson and Urbach [Howson & Urbach, 1989]: independent evidence types eliminate distinct confounders. Massimi's perspectival realism [Massimi, 2022] maps directly onto the perspectival view, with the important distinction that perspectival evidence does not entail perspectival truth.

Physics. The structural view's gauge-invariance requirement is the direct analogue of gauge symmetry in field theory: two weight configurations related by a computation-preserving transformation are physically the same mechanism, just as two vector potentials related by a gauge transformation describe the same electromagnetic field [Nakahara, 2003]. The stratified view borrows the renormalization group's central insight [Wilson, 1971, Goldenfeld, 1992]: effective descriptions change with measurement scale, and the interesting objects are the fixed points (stable strata) and relevant operators (structure that survives coarse-graining). Universality classes in physics correspond to mechanisms that appear identical at a given resolution despite differing microscopically.

Mathematics. The formalism axis is not merely notational. The choice of mathematical structure constrains what can be stated precisely and what can be proved. Grassmannian geometry [Nakahara, 2003] makes subspace proximity a metric statement; fiber bundles make gauge invariance a theorem rather than a heuristic check; Whitney stratification [Whitney, 1965, Golubitsky & Guillemin, 1973] gives a rigorous account of how mechanism identity changes across strata boundaries. Each formalism carries proof obligations that the corresponding informal claim does not.

Cognitive science and neuroscience. The localization–distribution spectrum (object view through stratified view) mirrors the longstanding debate between localizationism and distributed coding in systems neuroscience [Bechtel & Richardson, 1993]. The double dissociation methodology in neuropsychology—showing two functions can be selectively impaired independently—is the nearest analogue to the necessity and sufficiency tests used in the object view. Marr's tri-level account [Marr, 1982] organizes explanatory targets; the five axes here organize ontological commitments. Both are needed, and they are complementary: two researchers at the same Marr level can hold different mechanistic views.

8 Open Questions

Several questions are raised by the framework and not resolved:

1. *Process identity.* The identity criterion for the process view (same dynamical basin or

trajectory type) is underspecified relative to the object, subspace, and structural criteria. Formalizing process identity is an open problem.

2. *Cross-view promotion.* Under what conditions can an object-level result be promoted to a subspace-level or structural-level result? Section 7 gives preliminary conditions, but a systematic account of inter-view inference is missing.

3. *Completeness of the atlas.* The eight views are identified by reading the literature, not derived from the five axes. Whether there are important views not represented—a Marr computational-level view [Marr, 1982], a representational-content view, a population-level statistical view—is an open question.

4. *Sufficiency of the five axes.* The five axes may be too many or too few. Evidence and formalism are tightly constrained by ontology and identity (the determination chain), raising the question of whether they are genuinely free axes or downstream consequences. Conversely, the framework omits a normative dimension—it diagnoses coherence but not adequacy for a given purpose. Whether adequacy is a sixth axis or a meta-level concern about the framework is open.

5. *View-independence of the framework.* The framework itself is presented as view-neutral, but the formalization (equivalence relations, typed evidence) may privilege views whose identity criteria are crisp. Whether the framework adequately represents views with inherently fuzzy identity (perspectival, process) is open.

6. *Views and safety-relevant inference.* Safety applications (monitoring, alignment auditing) require mechanistic claims that are robust to adversarial distribution shift. What minimum view supports safety-relevant inference? The object view is likely insufficient—a head identified as safety-relevant on one distribution may not generalize. The structural or stratified views may be necessary, but their evidential demands are currently impractical at scale.

7. *Implicit view commitments of existing programs.* The monosemanticity program [Bricken et al., 2023, Templeton et al., 2024] implicitly assumes that individual SAE dictionary elements are the right ontological unit (object view) with feature-level identity. The causal abstraction program [Geiger et al., 2021, 2024] assumes role-level identity via interchange intervention. Making these commitments explicit would clarify what each program can and cannot establish.

9 Conclusion

Mechanistic interpretability is not only a search for circuits. It is a search for the right kind of object to call a mechanism. The field already uses multiple mechanistic views, typically implicitly, and making those views explicit improves both the precision of claims and the appropriateness of evidence.

The central contribution is definitional: a mechanistic view is a five-component structure $(O, \sim, E, F, T)$, and every mechanistic interpretability result assumes one, whether or not it is stated. Coherence between the components is not automatic: incoherent views generate systematically unsatisfiable evidential demands (Proposition 2.1) and unresolvable underdetermination (Proposition 2.2). The determination chain—ontology implies identity implies formalism—explains why most coherence violations involve mismatches between these first three axes.

The atlas of eight views—object, role, subspace, structural, process, stratified, perspectival, instrumental—covers the main options currently in use. Each view is appropriate for some questions and inappropriate for others. The paper is pluralist: the goal is not to collapse this plurality into a master ontology but to make view choice explicit, justifiable, and revisable. Methods are discussed only in their relation to views, not evaluated on their own technical merits. The induction head result remains the canonical illustration of what convergence across views looks like: object, role, and process views all point to the same entity, which is why the result supports claims that single-view results cannot.

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A Coherence Violations: A Diagnostic Checklist

The following patterns indicate a coherence violation in the view underlying a claim.

(1) **Ontology–identity mismatch.** The ontology declares mechanisms to be concrete components, but the identity criterion is functional (two different components can be the same mechanism). Resolution: adopt the role view, which has functional identity built in, or restate the claim as object-level and drop cross-model generalization.

(2) **Evidence out of range.** A structural or subspace claim is supported only by activation-patching evidence. Resolution: identify the view that patching evidence actually supports (object view) and state the structural or subspace claim as a hypothesis requiring further evidence.

(3) **Unrestricted alignment.** A subspace claim is supported by DAS/IIA without specifying the alignment class or reporting subspace stability. By the Sutter result [Sutter et al., 2025], high IIA under unrestricted nonlinear alignment is consistent with any causal structure. Resolution: restrict the alignment class to linear maps or transport-respecting maps and report the eigenvalue gap.

(4) **Process claim from checkpoint evidence.** A claim about how a mechanism formed or why a behavior emerged at a certain scale is supported only by final-checkpoint analysis. Resolution: either restate as a static claim or obtain training-dynamics evidence.

(5) **Object-to-role slide.** A component is identified by ablation and immediately labeled with a role without an independent test of the role specification. Resolution: operationalize the role prediction and test it on held-out constructions.

B Full Axis Assignments

Table 3 repeats the five-axis assignments from Table 1 and adds the characteristic failure mode for each view.

C Open Problems in Mechanistic Interpretability

We surveyed nine sources spanning hundreds of individual open questions in mechanistic interpretability [Sharkey et al., 2025, Nanda, 2022, Balesni et al., 2024, Schmidt Futures, 2026, Steinhardt, 2026, Williams et al., 2025, MIB, 2025] and distilled 24 distinct problem types. Each maps onto specific view-level confusions, ceilings, or under-explored territories. Of the 24: 11 are **dissolved** (the question disappears when you name the view), 2 are **reframed** (the framework clarifies what evidence would answer it), and 11 are **scoped** (the framework identifies which view owns the problem).

C.1 Dissolved Problems

These questions are ill-posed: they disappear once the view is named.

Decomposition identity. SAE features change with dictionary width—which decomposition is "real"? Under the object view, this is a problem: the enumeration depends on resolution. Under the subspace view, it is expected: the *subspace* spanned by a cluster of split features is what is real; how you decompose it is gauge freedom. SAEs finding features in noise [Bricken et al., 2023] is a perspectival-view prediction: decompositions are partly method artifact, requiring cross-method validation.

SAE "true" features. "Do SAEs recover the true features?" presupposes an object-view ontology in which discrete features are the fundamental units. Under the subspace view, the question becomes whether the subspace converges across widths and seeds—measurable via Grassmannian distance.

Superposition. "How do models represent more features than dimensions?" is a problem under the object view (feature entanglement to decompose), a geometric fact under the subspace view (expected overlap), and a presupposition under the perspectival view (it assumes discrete features exist to be "superposed").

Circuit disagreements. ACDC, EAP, and activation patching find different circuits for the same task. Under the object view, at most one can be correct. Under the role view, each method identifies components that play the relevant role under its own intervention semantics—they need not agree, and disagreement is diagnostic of different implicit identity criteria.

Unit of analysis. Are attention heads, neurons, or subspaces the right unit? The unit depends on the view: object view uses architectural components, role view uses functional roles, subspace view uses causal subspaces.

Probe features. "Does the model use feature X?" means four different things: decodable (instrumental), more than artifact (perspectival), causally active (subspace), or functionally necessary (role).

Cross-model identity. "Same features across models" requires specifying the identity criterion. Same top-activating examples (instrumental) is weak. Same ablation effect (object) is architecture-dependent. Same subspace (subspace) is measurable via Grassmannian distance. Same gauge orbit (structural) is the strongest criterion. Most universality claims use instrumental evidence for structural conclusions.

DAS in random models. Unconstrained DAS achieving high IIA on random models [Sutter et al., 2025] is a perspectival-view prediction: unrestricted nonlinear alignment always finds structure, so what it finds in a real model needs stronger validation.

C.2 Reframed Problems

These questions remain empirically open, but the framework clarifies what evidence would answer them.

CoT faithfulness. "Is chain-of-thought faithful?" is four questions: does CoT predict behavior (instrumental), converge with other methods (perspectival), describe real functional roles (role), or map onto the computational graph (structural)? The safety-relevant notion requires cross-method convergence that no one has demonstrated.

World models. Current evidence that models encode state predicates linearly is subspace-level (the encoding exists). Whether these encodings form a causal world model—in which the model's internal causal graph mirrors the world's—is a structural claim requiring evidence of causal relationships between the encoded variables, not just their individual existence.

C.3 Scoped Problems

These are real and structural, arising from the limits of the view being used.

Intervention limits. Ablation-based methods face three Object-view ceilings: reconfiguration (the remaining network compensates), interaction effects (single-component ablation misses synergies), and side effects (ablating multi-functional components conflates necessity with general stability). Moving to the subspace view (orthogonal interventions) or structural view (gauge-invariant properties) partially resolves each.

Linear assumption. Circular and manifold representations challenge the assumption that concepts are encoded as linear directions. This is a subspace-view ceiling: the current formalism assumes linear subspaces, but the evidence increasingly shows nonlinear structure.

Safety evidence gaps. Prominent safety findings (refusal direction ablation, representation engineering for truthfulness) are instrumental evidence: they establish that a direction is a sufficient behavioral lever, not that it is the mechanism. Safety monitoring requires at minimum object evidence (necessity plus sufficiency) and ideally subspace evidence (stability across distributions).

Validation methodology. Three foundational threats to MI evidence: validation circularity (testing on the discovery distribution), interpretability illusions (distribution-dependent feature labels), and lack of random baselines (arbitrary directions receive plausible labels). All are predicted by the perspectival view and resolved by requiring convergent evidence across methods and distributions.

Training dynamics. Most MI analyzes static snapshots, but mechanisms form and can be destroyed during training. Fine-tuning silently destroys circuits; adversarial training can relocate mechanisms; phase transitions create mechanisms suddenly. These belong to the process view, which uniquely provides evidence for cross-seed reliability and robustness to continued training.

Resolution and completeness. "How much do we understand?" is ill-posed without specifying resolution. Known circuits explain nearly everything at coarse granularity but effectively nothing at fine granularity. This is stratified-view territory: the interesting objects are the stable strata and the structure that survives coarse-graining.

Deceptive alignment. Detecting deceptive alignment requires evidence far above what current methods produce. Instrumental evidence (steering vectors, probes) is structurally insufficient: competent deception is behaviorally identical to alignment. The minimum standard is structural evidence (gauge-invariant deception circuit) combined with process evidence (formation trajectory during training).

C.4 Resolution Summary

Method	Ontology	Identity	Evidence	Formalism	Target
Activation patching	Object	Component overlap	Activations	Directed graph	Task circuit
Path patching	Object	Component overlap	Activations	Directed graph	Task circuit
ACDC	Object	Component overlap	Activations	Directed graph	Task circuit
EAP	Object	Component overlap	Act. + Wt.	Directed graph	Task circuit
Ablation	Object	Component overlap	Activations	Directed graph	Importance
DAS / IIA	Role	Role equivalence	Activations	Subspace (borrowed)	Concept
Causal scrubbing	Role	Role equivalence	Activations	Causal graph	Alignment
Linear probing	Role	Role equivalence	Activations	Linear classifier	Detection
SAE features	Object	Component overlap	Activations	Dictionary	Feature catalog
Logit / tuned lens	Object	Component overlap	Activations	Linear proj.	Layer readout
SVD of weights	Subspace	Subspace prox.	Weights	$\mathrm{Gr}(k, d)$	Decomposition
Composition scores	Structural	Gauge orbit	Weights	Fiber bundle	Info-flow bound
Factorization	Subspace	Subspace prox.	Weights	$\mathrm{Gr}(k, d)$	Decomposition
Factor EAP	Subspace	Subspace prox.	Wt. + Act.	$\mathrm{Gr}(k, d)$ + graph	Task circuit
AGOP	Process	Basin membership	Dynamics	Dynamical sys.	Trajectory

Table 2: Five-axis classification of common interpretability methods. Most methods are Object view with activation evidence. DAS uses a subspace *parameterization* but Role *identity*: it validates by IIA success, not Grassmannian distance. Factor EAP is the only method combining Subspace ontology with both weight and activation evidence.

View	Ontology	Identity	Evidence	Formalism	Target	Failure mode
Object	Concrete part	Component overlap	Ablation, patching	Directed graph	Specific behavior	Role inflation; distribution sensitivity
Role	Functional role	Role equivalence	Role-specific causal tests	Functional decomp.	Functional class	Role proliferation; underdefined roles
Subspace	Causal subspace	Same projector	DAS/IIA (linear)	$\mathrm{Gr}(k, d)$	Representational variable	Sutter vacuousness; SAE conflation
Structural	Gauge-invariant structure	Gauge orbit	Holonomy, composition	Fiber bundle	Computation class	False gauge invariance; arch. prior confound
Process	Formation trajectory	Trajectory type	Checkpoints, knockouts	Dynamical system	Mechanism origin	Process–circuit conflation; training overfitting
Stratified	Stratum point	Stratum + local equiv.	Dimensionality, rank	Whitney strat.	Resolution-relative	Stratum misidentification from method artifacts
Perspectival	Method projection	Cross-method coherence	Multi-method robustness	Measurement algebra	Method-relative	Anti-realist slide; perspectival conflation
Instrumental	Predictive model	Predictive equiv.	Forecast, intervention	Model theory	Behavior prediction	Safety-relevant overreach

Table 3: Full axis assignments and characteristic failure modes.

Problem	Status	Key views
Decomposition identity	Dissolved	Object vs. Subspace
SAE "true" features	Dissolved	Object vs. Subspace
Superposition	Dissolved	Object vs. Subspace vs. Perspectival
Circuit disagreements	Dissolved	Object vs. Role
Unit of analysis	Dissolved	Object vs. Role vs. Subspace
Probe features	Dissolved	Instrumental vs. Subspace vs. Role
Cross-model identity	Dissolved	Object impossible, Subspace possible
DAS in random models	Dissolved	Perspectival diagnosis
CoT faithfulness	Reframed	Instrumental vs. Structural
World models	Reframed	Subspace vs. Structural
Intervention limits	Scoped	Object ceiling
Linear assumption	Scoped	Subspace ceiling
Safety evidence gaps	Scoped	Instrumental floor
Validation methodology	Scoped	Perspectival diagnosis
Training dynamics	Scoped	Process territory
Resolution/completeness	Scoped	Stratified territory
Deceptive alignment	Scoped	Structural + Process required

Table 4: Resolution status of 24 open problems (17 distinct types) across 9 sources. Dissolved: the question disappears when you name the view. Reframed: the framework clarifies what evidence would answer it. Scoped: the framework identifies which view owns the problem.